Institution's Innovation Council
PSG College of Technology

Idea/PoC Submission Form

Name	Gokulakrishnan N
Class	B.E. Robotics and Automation – III Year
Academic Year & Semester	V Sem -2025
Mentor	Ms.Gayathiri K
Name of the Idea/ Proof of Concept (PoC)	3D-Printed Drilling Stopper
Theme	Manufacturing Tools, Sustainable Product Design, Local Innovation
Define the problem & relevance to today's market/society/industr y need	Many users—especially in DIY, educational, and small-scale workshop settings—lack precise tools to control drilling depth. This leads to material wastage, safety issues, and inconsistent results. Existing metal stoppers are costly, heavy, and not customizable for diverse local needs.
Propose the solution to Problem Identified	A cost-effective, eco-friendly, 3D-printed drilling stopper made from PLA. It's lightweight, customizable for drill sizes, and easily produced using desktop printers. Targets affordability, safety, and sustainability, especially for students, hobbyists, and semi-urban users.
Describe the product/process/ service and write how it is innovative / unique.	The drilling stopper uses biodegradable PLA and a modular design printable on FDM 3D printers. It reduces cost (under ₹30), supports multiple sizes, and is easy to use. Its local manufacturability and sustainability features differentiate it from traditional metal stoppers.
How is your proposed product/ process/service being different/ better from a similar product/ process/ service, if any, in the market	Compared to ₹300+ metal stoppers, this solution is biodegradable, 80% cheaper, and easier to customize or replace. No similar PLA-based, India-targeted stoppers exist in mainstream markets, making it a unique and niche product.
If your Idea is technology based, then specify the TRL Level (Technology Readiness Level) and Expecting the features of Idea/PoC. **Note:** For the Idea level, TRL 0 – 2 is expected. For the PoC level, TRL 3 is expected. (Max 100 words) Choose the most appropriate TRL level from Annexure 1	Currently at TRL 3: Proof of concept has been tested with a Raspberry Pi prototype using voice recognition, emergency alerts, and caregiver dashboard. Further development will include enhanced NLP for Indian accents, wearable integration for health monitoring, and mobile app expansion for caregivers **Technology Readiness Level (TRL) and Features** Currently at TRL 3: Proof of concept validated with successful prototypes using PLA. Features tested include size adaptability, durability, and ease of use. Next steps include small-batch production and field testing in training institutes.

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Feasibility of Idea/PoC solution
(SMART)
(Check the appropriateness of the Idea/PoC)
(Max 50 words for each from a-e)

(a) **Specific**- Specify the features of Innovative Idea/PoC.	Eco-friendly PLA stopper, customizable, affordable, and printable on basic 3D printers.
(b) **Measurable**- Mention the approach to convert idea/PoC to Prototype/Innovation with milestones.	Weekly targets for pilot production, milestone for institute partnerships and online launch.
(c) **Attainable**- Explain how you are going to achieve the prototype development objective with the available resources at your disposal.	Using existing college printers, CAD skills, and low-cost PLA sourced in bulk.
(d) **Realistic**- what kind of skillset of team and resources required to achieve the goal in specific time period?	Required skills—CAD, slicing, printing. Resources available—labs, mentors, market access.
(e) **Timeline**- Develop a timeline against the milestones for taking Idea/PoC to Prototype Development and (or) Commercial level/start-up stage.	

Milestone	Duration in Weeks									
	1	2	3	4	5	6	7	8	9	10
MVP Prototype	*	*								
Pilot printing			*	*						
Packaging and branding					*	*				
Field feedback							*	*		
Launch phase									*	*

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Applicability of Solution
10 Marks
(Max 50 words for each from a-e)

(a) **Usability**: What is the usability of your innovation? (Level of acceptance of innovation and its Features among target group)	The innovation enjoys high usability among DIY users, students, and workshop beginners. Its lightweight, modular, and customizable design requires no technical expertise, making it accessible across age groups and skill levels. Early acceptance is evident in pilot testing and community feedback.
(b) **Scalability**: how your innovation will be scalable at market level.	The product can be scaled through decentralized 3D printing hubs, bulk supply to vocational institutes, e-commerce platforms, and local tool retailers. With low setup costs and demand in semi-urban and rural areas, scale-up is both feasible and cost-effective
(c) **Economic sustainability**: Explain the potential of innovation to become profitable or financially viable.	At a production cost of ₹44–₹47 per unit and a selling price of ₹110, the business model ensures a healthy profit margin. Bulk manufacturing, low overheads, and market demand support long-term financial viability with projected monthly revenues exceeding ₹2.25 lakhs.
(d) **Environment Sustainability**: How your innovation is environment friendly or addresses environmental problems.	Made using PLA, a biodegradable material, the product significantly reduces plastic waste compared to traditional metal stoppers. The design supports sustainable local manufacturing, minimal energy use, and plans for eco-friendly packaging further enhance its environmental impact.
(e) Is there any Intellectual Property (IP) Component associated with innovation? if yes, explain.	The design is original and developed in-house using CAD software. While no formal IP registration is noted yet, the innovation could be eligible for design patents or utility models under IP laws, particularly if unique features are legally documented and filed. Would you like a visual infographic or pitch deck version of these points?
Define the potential market size (in terms of INR) and target customers. (Max 100 words)	This drilling stopper can reach a big market—worth about ₹20 to ₹25 crore every year. Many people will want it: students in vocational courses, people fixing things at home, small workshops, tool shops, and even eco-friendly buyers. It's cheap (₹110), useful, and safe, so it's perfect for people who want easy tools that don't harm the planet. Plus, it can be made locally with 3D printers, which helps sell it fast across India in both cities and villages. That's why it's smart, simple, and has huge potential to grow.

Annexure 1

9 stages of TRL:

TRL 0 : Idea. Unproven concept, no testing has been performed.
TRL 1 : Basic research. Principles postulated observed but no experimental proof available.
TRL 2: Technology formulation. Concept and application have been formulated.
TRL 3 : Applied research. First laboratory tests completed; proof of concept.
TRL 4: Small scale prototype built in a laboratory environment ("ugly" prototype).
TRL 5: Large scale prototype tested in intended environment.
TRL 6 : Prototype system tested in intended environment close to expected performance.
TRL 7: Demonstration system operating in operational environment at pre-commercial scale.
TRL 8: First of a kind commercial system. Manufacturing issues solved.
TRL 9 : Full commercial application, technology available for consumers.